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An Interior Decoration System via Virtual Reality and Artificial Intelligence

Xia Meng¹, Saba Ghazanfar Ali¹, Ping Li^{2,*}, Anum Masood¹, Bin Sheng^{1,*}, and Jie Ren³

1. Department of Computer Science and Engineering, Shanghai Jiao Tong University, Shanghai, China

2. Faculty of Information Technology, Macau University of Science and Technology, Macau, China

3. Shanghai University of Sport, Shanghai, China

*corresponding authors

Email: pli@must.edu.mo, shengbin@sjtu.edu.cn

Abstract—In this paper, we proposed an interior design system. We have implemented the scene selection function and the house type drawing function to get the apartment type. After getting the basic apartment type, we have also provided some other basic decoration functions such as furniture placement, furniture conversion, material conversion, and light switch. These functions are operated by mouse clicking and keyboard control. In addition, we have added some AI modules to provide an additional assistant. Through the recognition of the picture, the texture can be trained, and the ideal texture has been obtained. The implementing environment of our design system is UE4, and the AI algorithm was written in Python and tensor flow.

Index Terms—Interior design, AI, CycleGAN.

I. INTRODUCTION

The evolution of virtual reality (VR) technologies has been used in numerous fields of social life. And it promotes the traditional architectural design and interior decoration design [1]–[7]. This new design form can display the design more directly in front of people to meet people's requirements for decoration. It promotes communication between customers and designers and enables the design to better meet the needs of customers. This article mainly introduces the interior design system that includes furniture placement, real-time conversion of materials, and also explains the virtual reality technology in the system. UE4 (unreal engine 4) is a game engine developed by epic games. Using the UE4 engine, we developed the functions required for the interior design described in our system. About AI [8]–[11], we used CycleGAN [12]. Since the texture making process has always been a trouble to new beginners. In this paper, we have tried to implement CycleGAN in UE4 to transform texture style by means of CycleGAN's strong ability of style translation. Basically, the principle of Generative Adversarial Networks (GANs) is to create an adversarial loss that is used for the differentiation of produced images and real images. But all existing works require paired training examples, which refer to the concept of image-to-image translation. The main work of CycleGAN

is capture special features of an image group and assuming how these features can be interpreted into alternative image group, not in the presence of matching training examples which is also its biggest advantage. These are the following main contributions for our work:

- **Incorporate AI method:** Incorporated AI methods to achieve certain effects. The specific content is described later in this article.
- **Realize the choice and autonomous drawing of house type:** In the system of this article, we have implemented two methods of house type design which selects the existing house type and drawing a new one.

II. RELATED WORK

-Intelligent design CycleGAN The idea of image-to-image conversion is to pay consistence model for one input and output training image pair. These type of ideas have applied in different errands for producing photographs from sketches, features, or semantic designs also these type of methods need pairing. As a result of the short comings of image-to-image translation, different methods used to tackle the unpaired image-to-image translation. Like [13] use cross-model scene networks for weight-sharing problems to acquire a mutual illustration diagonally. Another line of simultaneous work inspires the input-output distribute the convinced content features if they are different in styles. The basic concept relates the idea of without pairing image-to-image translations are cycle consistency, that use transitivity as a technique to normalize organized data. Recently, higher order cycle consistency has applied by many types of research. In [14], [15], practices sequence reliability loss for consuming transitivity to managed CNN training. [16], independently uses a same purpose for unpaired image-to-image translation.

-About interior decoration The combination of VR and design is very significant and a lot of relevant work have already existed. Because VR technology is very important to Computer Integrated Engineer System, the VRML-Java based on VR technology is used to develop the visual design system [17]. Different people design interior decoration system by the 3D scene modeling technique [18]. When discussing the application of VR in interior decoration, combined with actual development experience, related techniques such as view scene

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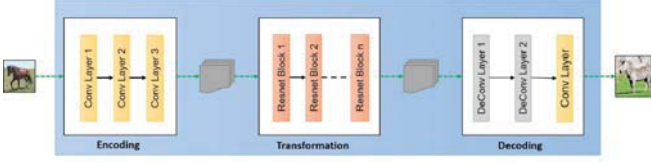


Fig. 1. Generator

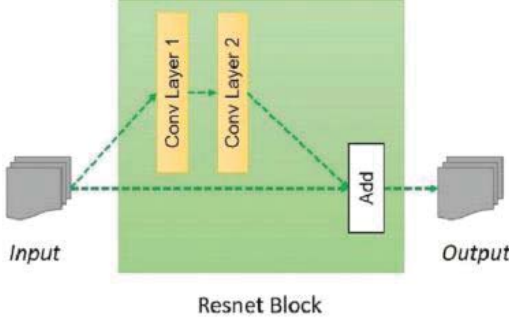


Fig. 2. Transformer

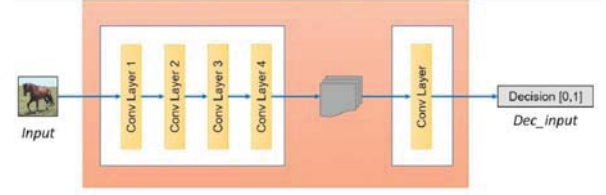


Fig. 3. Discriminator

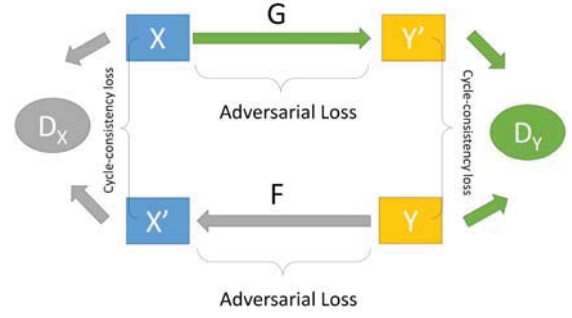


Fig. 4. Full objective of CycleGAN

modeling, real-time rendering, roaming and sound effects under the virtual environment are described [19]. Thus, our work has been integrated on the premise of combining the above-mentioned content, and incorporated AI technology, adding certain challenges while applying traditional technologies.

III. APPROACH OVERVIEW

A. AI Methodology

-Generator Two generators work in the same way, take Generator X as an example. Generator X consists of 3 parts: encoder, transformer and decoder.

-Encoder: For simplicity, the input has been fixed as [256,256,3]. By using convolution network, we first extract features. A convolution network takes an image as input, then uses filters in the image to extract features each time.

-Transformer: This part is to transform feature vectors of image with domain X to domain Y . For this, we have used 6 layers of resnet blocks. Resnet block can be summarized as follows, which contains a direct channel from the input to the output and two convolution layers.

-Decoder: Decode and encode is opposite twill rebuild an image from to each other and image was build by using feature vectors gained before, and this is done by applying three de-convolution layers which use reversed parameters of encoding step.

-Discriminator: Discriminator also needs to extract features, so it is basically a convolution network. So the next work is to decide whether they are matching with specific area or not. For 1D output, a final convolution layer has added.

-Loss: To meet our goal, the loss function must satisfy: Discriminator X should be trained because it is necessary for

images from domain X close to one. So, discriminator X minimize as $(DiscriminatorX(x)-1)^2$. Also, since discriminator X differentiate between generated and original images, it predict zero for the images that generates by generator. Discriminator X minimize the $(DiscriminatorX(GeneratorY \rightarrow !X(y)))^2$. Moreover, generator X should gradually sure the discriminator Y for authenticity of its output images. Because the recommended discriminator Y of output images is near to 1. Thats why, generator X minimize the $(DiscriminatorY(GeneratorX!Y(x)) \rightarrow 1)^2$. And the most important cycle loss captures previous image with the usage of different generator and the difference of images should be as small as possible.

B. Interior Decoration Methodology

We use UE4 software. The interior decoration design system has rich functions. Users could use this system to draw the house type with any size and any structure. The user also could design the house with rich furniture. The user could place the furniture, move the furniture, and rotate the furniture. The user could also add the light to the room.

- Material replacement of the floor wallpaper is achieved.
- Furniture items are selected from the library for placement and can be rotated and scaled.
- User could move in each room.
- User could learn the top view of the overall browsing unit.
- Save and read design.
- User could edit the type of house by drawing units and adding doors or windows.
- Increase the animation of furniture.
- Increase the function of analog lighting.



Fig. 5. Interior Decoration Design System

IV. AI PART

In this part, we will introduce the AI module for our system.

A. Motivation

During the process of implementation, the most head-scratching part comes from the texture and maps. We learned to transform some 3D's max models into UE4, there were always some incompatibility errors. And as a result, the transformed model only contains the shape information with gray colors which means all the texture and maps were lost. We have found that most of the models downloaded from the internet used the V-ray material. Although V-ray is very powerful and realistic, unfortunately, UE4 does not support this kind of rendering method yet. Afterwards, we tried to convert the V-ray material into the normal material. The texture information is kind of retained, but there were still some color deviation or other strange problems. Besides, sometimes when we import models into UE4, we have to attach those textures or maps to models manually, which is also very time-consuming. Even if we convert models successfully, the model's appearance is fixed. If you want to change models color or texture, you need to either prepare many variegated models or make other textures by yourself. Both ways are very time consuming and space-consuming.

B. Proposed Method

Here comes the idea of making textures with the help of AI, specifically (GAN). Image transformation is a very popular topic nowadays, especially with the help of neural networks. Most of the image translation is about one-to-one mapping, which in other words, used pairwise images to train the model. However, such pairwise images are not always available. And most of the time, we are only provided with groups of pictures of a similar feature. CycleGANs goal is to implement an unpaired image translation. This translation learns the mapping $G : X \rightarrow Y$ (X, Y two groups of images of X, Y source and target domain) such that in ideal case, the generated image $G(X)$ has the same distribution of images in Y . And in this way, the generated image has some features of images in Y .

The basic idea of implementation is to use adversarial networks. The generator network attempts to generate images from the desired distribution, also the discriminator network needs to distinguish whether an image is authentic from the target domain or a not genuine generated from source domain. And by competing with each other, finally we will

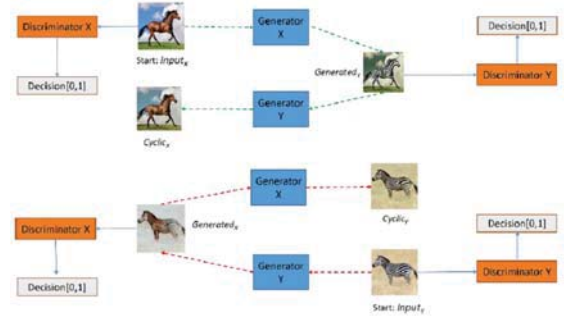


Fig. 6. Pipeline of CycleGAN



Fig. 7. Horse to Zebra

get the desired generator. Since there is huge measurements, for mapping a network in the input images are able to satisfy such distribution, in this way we can hardly get the desired output images. And there are occasions when the model collapses, and inputs maps the output images and the optimization process fails for the making of progress. As a result, another constraint needs to be added which is the cycle-consistency. To transform image from source distribution to target and then vice versa, The samples has been received by source distribution. And with the help of this, successful transformation can be achieved. We have used CycleGAN (a variational version of GAN) to train some funny models. In brief, CycleGAN takes two datasets with different styles as input, and it will learn the transformation pattern between them. For example, by providing a horse image set and a zebra image set, CycleGAN will learn to transform a horse in an image into a zebra without changing anything else, or the other way around. The advantage of CycleGAN lies in that you do not need to provide pairwise images to train which means the horse dataset can contain any kind of casual horses and the zebra dataset can contain any kind of casual zebras and they do not have to match each other. This advantage of CycleGAN is very suitable for texture transform because there are textures with different UV maps and textures with different light features or metallically which makes it difficult to find pairwise datasets. Thus, with the help of CycleGAN, we can prepare a standard dataset with gray colors textures and different target datasets, such as wood material dataset or grass material dataset. Then, we can use CycleGAN to train different conversion models by feeding a different combination of datasets and import these models into UE4 afterwards. The easiest version of texture transform is just the transformation of maps. The computational detail and the network structure

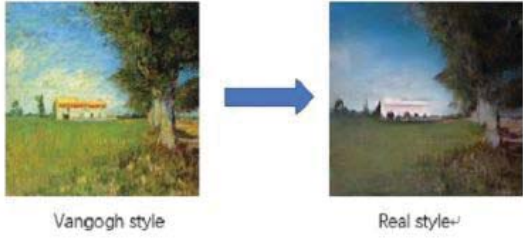


Fig. 8. Vangogh style to Real style

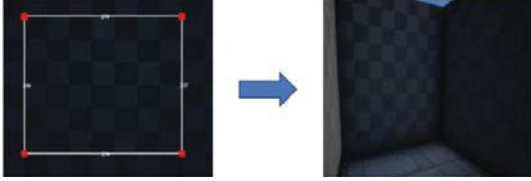


Fig. 9. House Drawing Function

is shown below:

- Domain: X, Y .
- 2 maps $G : X \rightarrow Y$ and $F : Y \rightarrow X$.

-Adversary Loss: An adversarial loss has been applied together with mapping functions $G : X \rightarrow Y$ with discriminator D_Y , the adversarial loss is:

$$\zeta_{GAN}(G; D_Y; X; Y) = E_{y \sim P_{data}(y)}[\log D_Y(y)] + E_{x \sim P_{data}(x)}[\log(1 - D_Y(x))]$$

G generates images like $G(x)$ which is similar to images of domain Y , and D_Y differentiate among the $G(x)$ translated images and the samples of y . G reduce the adversary D which tries to increase it. The same adversarial loss is introduced $F : Y \rightarrow X$ and D_X for the mapping function and for discriminator respectively. **-Cycle Consistency Loss:** Adversarial loss will not ensure that the learned function maps input separately x_i with the final output y_i . For mapping functions, Every images x from domain X should be consistent and translation cycle of image x backward to the original image, i.e.

$$x \rightarrow G(x) \rightarrow F(G(x)) \approx x$$

Similarly

$$y \rightarrow F(y) \rightarrow G(F(y)) \approx y$$

We could incentive cyclic consistency loss:

$$\zeta_{cyclic}(G; F) = E_{y \sim P_{data}(y)}[\|F(G(x)) - x\|_1] + E_{x \sim P_{data}(x)}[\|G(F(y)) - y\|_1].$$

-Full Objective:

$$\zeta(G; F, D_X, D_Y)$$

$$= \zeta_{GAN}(G, D_Y, X, Y) + \zeta_{GAN}(F, D_X, Y, X) + \lambda \zeta_{cyc}(G, F)$$

where λ control 2 objectives that solved

$$G^*; F^* = \operatorname{argmin}(G, F) \max(D_X, D_Y)$$

$$\operatorname{mathcal{L}}(G, F, D_X, D_Y).$$

C. What We Have Done

We have explored the possibility of combining tensor flow and CycleGAN with UE4. Meanwhile, we have implemented some pre trained models in UE4 to show the amazing effect

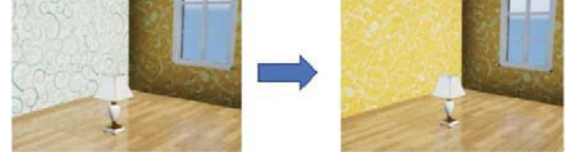


Fig. 10. Material Conversion Function

of CycleGAN just as shown below. In detail, we have hung a painting in our scene as a decoration. By selecting it and press the F key, our program will automatically apply a transformation model on it. After the transformation work is done, the painting on the wall will be replaced by the regenerated one.

V. FUNCTION PART

The interior decoration design system has rich functions. Users could use this system to draw the house type with any 5 and any structure. The user also could design the house with rich furniture. We will introduce the function of drawing the house type in Section A. We will introduce the function of material replacement in Section B. The function of placing furniture will be introduced in Section C. Furniture items are selected form the library for placement and can be rotated and scaled. We will introduce the function of Door or Window Settings in Section D. Finally, we will introduce the function of analog lighting in Section E.

A. House Drawing Function

House drawing is a very important feature. There are a lot of new units appearing every day. A large number of units means that we cannot list all the unit models for the user to choose. Therefore, the unit drawing function is very important. With this feature, users can draw any type of home they want, no longer limited by the diversity of the home. We create a three-dimensional floor plan by drawing a floor plan on a plane and converting it.

B. Material Conversion Function

The role of material conversion in interior decoration is very necessary. Many objects, such as walls, floors, ceilings, require a very close conversion of materials. This is because in the interior design, objects such as walls do not have as much flexibility as furniture. We cannot replace the different types of walls by replacing furniture. Therefore, we have implemented the material conversion function in the system. The user can complete the design of the wall by dragging the material in the material list onto the wall and other objects.

C. Furniture Placement Function

The furniture placement function is the core function of the interior decoration design system. After all, a good interior decoration, rich and varied furniture types, and unique furniture placement are inseparable. We have implemented functions such as placement, movement, rotation, and deletion of furniture. Users can place their favorite furniture in any position, and make the next judgment based on the effect.



Fig. 11. Furniture Placement Function



Fig. 13. Analog Lighting Function



Fig. 12. Door or Window Settings Function

D. Door or Window Settings Function

Doors and windows are also an important part of interior design. Our goal is not to design a closed house but a modern structure with a rich structure. Doors and windows are used as interfaces to connect externally, the setting of their position and the choice of style become very important. We added the ability to add doors and windows to the drawing units so that users can easily complete the design of this content.

E. Analog Lighting Function

A good interior design system must include lighting design. We added lighting furniture to the system and added lighting simulation capabilities to this furniture. The user can not only place the lighting like ordinary furniture, but also adjust the lighting switch to add lights.

VI. CONCLUSION

We implemented an interior decoration design system and an AI attempt in this system on UE4. The system basically satisfies the users needs in the traditional home improvement design, and the increased AI element makes the system more scalable. In the future, we will continue to improve our system. We can only transform some pictures in other word maps or textures of a furniture. But the showing effect of an object is not just about maps but also some metallicity and smoothness settings. So, in the future, we want to find a way to extract the whole material ball information and use self-defined coding scheme to train some more realistic and effective models. The interior decoration design system has also some limitations. The function of house drawing is not good. When using this function, User will find the walls does not fit perfectly. The number of material is small. We will improve the function of house drawing. We want the room has a more real appearance. We will also add more material to the system.

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